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Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

● Experiment No.: 1

- 1. Title:** Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.
- 2. Software/Tools Required:** Visual Studio Code (VS Code) Google Chrome/web browser, HTML, JavaScript.
- 3. Theory:** JavaScript is a scripting language used to make web pages interactive. It can be added in three ways:
 - Inline JavaScript: Written inside HTML elements.
 - Internal JavaScript: Written inside the `<script>` tag in the HTML file.
 - External JavaScript: Written in a separate `.js` file and linked to the HTML page.
 - Console methods like `console.log()`, `console.info()`, `console.warn()`, and `console.error()` are used for debugging and displaying messages.



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3. Output:

The screenshot shows a web browser window titled 'Shreya Maske - JS Exp 1'. The address bar shows the file path: `/Users/shreyamaske/jsexp1/webpage.html`. The page content is titled 'JavaScript Experiment 1' and contains three sections:

- Inline JavaScript (Welcome):** Displays 'WELCOME to JavaScript Lab Shreya!'.
- Internal JavaScript (Info):** Displays user information: Name: Shreya Maske, PRN: 24070521196, Branch: CSE, Year: Third Year.
- External JavaScript:** A placeholder text: 'Open Console to see console method outputs'.

The browser's developer console is open, showing the following messages:

- Output JavaScript Experiment 1 (js1.js:3)
- Information about Shreya Maske (js1.js:5)
- This is a warning message. (js1.js:7)
- This is an error message. (js1.js:9)
- Array(1) (js1.js:11)
 - 0: {Name: 'Shreya Maske', PRN: '24070521196', Branch: 'CSE', Year: 'Third Year'}
 - length: 1
 - [[Prototype]]: Array(0)
- Unsafe attempt to load URL `file:///Users/shreyamaske/jsexp1/webpage.html` from frame with URL `file:///Users/shreyamaske/jsexp1/webpage.html`. 'file:' URLs are treated as unique security origins. (webpage.html:1)

4. Result/Conclusion: The experiment was performed successfully. A webpage displaying a welcome message and user information was created using inline, internal, and external JavaScript, and console methods were demonstrated successfully.



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Case Study No.: 1

1. **Case Study Title:** Create a webpage for college department and link the student information page with it and write a java script to display “welcome to sit nagpur” in an alert box.

2. Case Study Program Code:

```
!DOCTYPE html>
<html>
  <head>
    <title>Shreya 24070521196</title>
    <!-- External JavaScript -->
    <script src="script.js"></script>
  </head>
  <body style="text-align:center;background-color: lightblue;">

    <h1>Symbiosis Institute of Technology, Nagpur</h1>
    <h2>Department of Computer Science and Engineering</h2>
    <hr style="border:2px solid red;"><hr style="border:2px solid red;">
    <h3 style="font-size:30px;">Student Information</h3>
    <a href="shinfo.html" target="_blank" style=font-size:30px;>Go to Student
Information Page</a>

    <br><br><br><br>
    <!-- Inline JavaScript -->
    <button onclick="alert('Welcome to SIT Nagpur')"style=font-size:30px;>
      alert
    </button>
    <br><br><br><br>
    <p style="font-size:30px;"><b>Open Console to see console.log()
output.</b></p>
  </body>
</html>
```



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```
<!DOCTYPE html>
<html>
<head>
  <title>Shreya Info 24070521196</title>
</head>
<body>
  <h1>Student Information</h1>
  <p><b>Name:</b> Shreya Maske</p>
  <p><b>PRN:</b> 24070521196</p>
  <p><b>Course:</b> B.Tech</p>
  <p><b>Branch:</b> Computer Science and Engineering</p>
  <p><b>Year:</b> Third Year</p>
  <br>
</body>
</html>
```

```
function welcomeExternal() {
  alert("Welcome to SIT Nagpur");
}

console.log("Welcome to SIT Nagpur");
```



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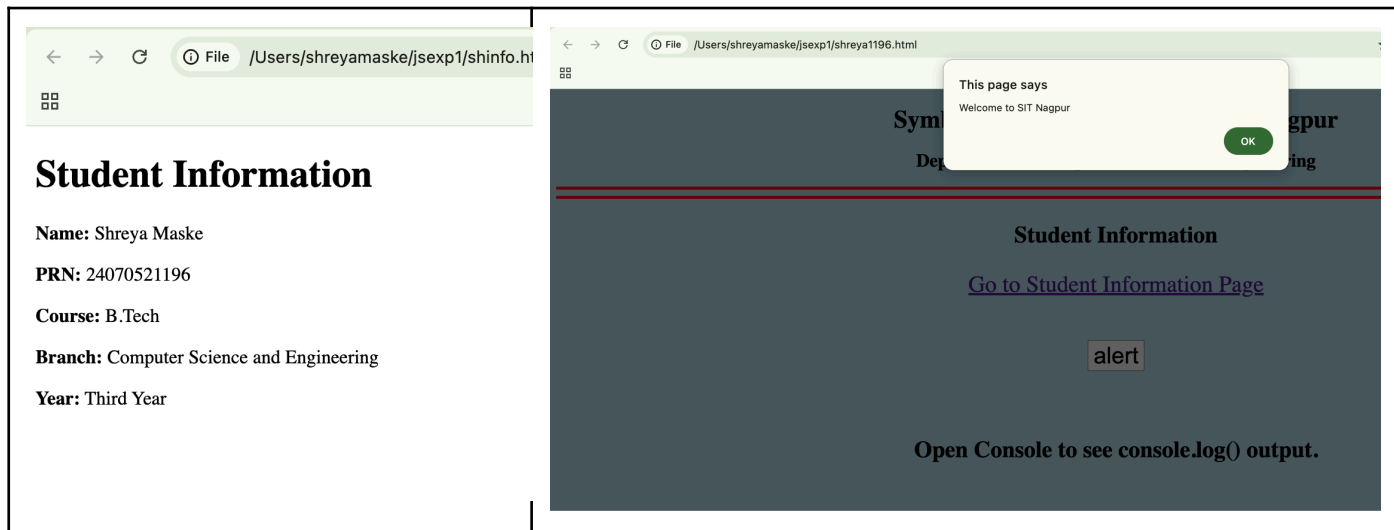
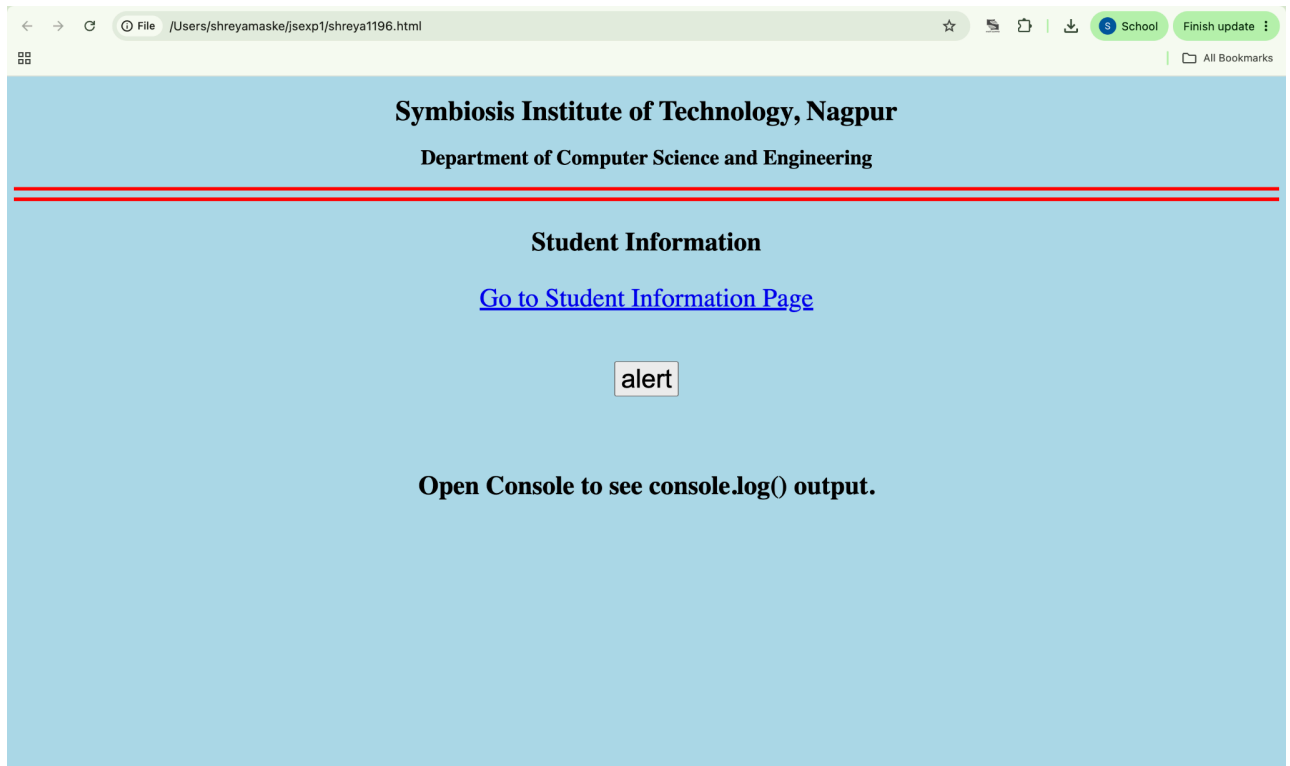
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3. Output:



3. **Result/Conclusion:** The college department webpage was created successfully. The Student Information page was linked to it, and JavaScript was implemented using inline, internal, and external methods. A welcome message was displayed in an alert box, and `console.log()` output was verified in the browser console.



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● Experiment No.: 2

1. Title: Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

2. Software/Tools Required: Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. Theory: `var` is function-scoped, `let` is block-scoped, and `const` is used for fixed values.

The `prompt()` function accepts user input, and `Number()` converts input into numeric values.

The program calculates the total amount, GST, and final bill using arithmetic operations.

An object stores all billing details together, while object destructuring extracts required values easily.

Template literals (``) are used to display formatted output with variable values using `$${}`.

The `document.write()` method displays the billing receipt on the webpage.

4. Output :

The output shows two screenshots of a web browser displaying a Billing Calculator interface. The first screenshot shows the 'Enter Customer Name' prompt with the text 'Shreya Maske' entered. The second screenshot shows the 'Enter Product Name' prompt with the text 'Microwave' entered. Both prompts have 'Cancel' and 'OK' buttons.



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The screenshot shows a web browser with two tabs: 'sec_apti_24-28: Attendance' and 'Billing Calculator'. The address bar shows the file path '/Users/pratham/jsexp2/shreya1196.html'. A dialog box titled 'This page says' is open, prompting the user to 'Enter Quantity :'. The input field contains the value '1'. There are 'Cancel' and 'OK' buttons at the bottom of the dialog.

The screenshot shows the same web browser with the 'Billing Calculator' tab active. The dialog box now prompts the user to 'Enter Price per Item :'. The input field contains the value '42000'. There are 'Cancel' and 'OK' buttons at the bottom of the dialog.

The screenshot shows the 'Shreya's Billing Calculator' web application. The title is 'Shreya's Billing Calculator'. Below it is the heading 'Billing Reciept' (note the typo). The receipt details are as follows:

- Customer: Shreya Maske
- Product: Microwave
- Quantity: 1
- Price per item: 42000
- Total: ₹42000
- GST (18%): ₹7560.00
- Final Bill: ₹49560.00**

5. **Result/Conclusion:** A billing calculator was successfully created using JavaScript. The program accepted user input, calculated the bill with GST, and demonstrated the use of var, let, const, template literals, and object destructuring effectively.



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Case Study No.: 2

1. **Case Study Title:** Develop the webpage for an online shopping website with using html,css and html tags and calculate bill,customer name,mobile no,invoice no,product no,quantity(kg),rate per kg,discount,GST,packing charges,payment mode(cash/card/UPI), Membership(yes/no).

2. Case Study Program Code:

```
shreya1196.html case2.html X
case2.html > html > body > script > bill
1 <!DOCTYPE html>
2 <html>
3 <head>
4 | <title>Online Shopping Bill</title>
5 </head>
6 <body style="background-color: ■ rgb(230, 218, 173);">
7 <h1>Shreya's Online Shopping Website</h1>
8 <script>
9 const GST = 0.18;
10 // Customer Details
11 var custname = prompt("Enter Customer Name:");
12 var mobile = prompt("Enter Mobile Number:");
13 var invoice = prompt("Enter Invoice Number:");
14 var product = prompt("Enter Product Name:");
15 var quantity = Number(prompt("Enter Quantity (Kg):"));
16 var rate = Number(prompt("Enter Rate per Kg:"));
17 var discount = Number(prompt("Enter Discount (%):"));
18 var packing = Number(prompt("Enter Packing Charges:"));
19 var payment = prompt("Enter Payment Mode (Cash/Card/UPI):");
20 var membership = prompt("Membership (Yes/No):");
21 // Calculations
22 let total = quantity * rate;
23 let discountAmt = total * discount / 100;
24 let subtotal = total - discountAmt;
25 let gstAmt = subtotal * GST;
26 let finalBill = subtotal + gstAmt + packing;
```



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```
27 // Object
28 const bill = {
29     customer: custname,
30     mobile,
31     invoice,
32     product,
33     quantity,
34     rate,
35     discount,
36     packing,
37     payment,
38     membership,
39     total,
40     discountAmt,
41     gstAmt,
42     finalBill
43 };
44 // Destructuring
45 const {
46     customer,
47     mobile: mob,
48     invoice: inv,
49     product: pname,
50     payment: payMode,
51     membership: member,
52     gstAmt: gst,
53     finalBill: amount
54 } = bill;
55 // Output
56 document.write(`<h2>Shopping Invoice</h2>`);
57 document.write(`<hr>`);
58 document.write(`<p><b>Customer Name:</b> ${customer}</p>`);
59 document.write(`<p><b>Mobile No:</b> ${mob}</p>`);
60 document.write(`<p><b>Invoice No:</b> ${inv}</p>`);
61 document.write(`<p><b>Product:</b> ${pname}</p>`);
62 document.write(`<p><b>Quantity:</b> ${quantity} Kg</p>`);
63 document.write(`<p><b>Rate per Kg:</b> ₹${rate}</p>`);
64 document.write(`<p><b>Total:</b> ₹${total.toFixed(2)}</p>`);
65 document.write(`<p><b>Discount:</b> ₹${discountAmt.toFixed(2)}</p>`);
66 document.write(`<p><b>GST (18%):</b> ₹${gst.toFixed(2)}</p>`);
67 document.write(`<p><b>Packing Charges:</b> ₹${packing.toFixed(2)}</p>`);
68 document.write(`<p><b>Payment Mode:</b> ${payMode}</p>`);
69 document.write(`<p><b>Membership:</b> ${member}</p>`);
70 document.write(`<h2>Final Bill: ₹${amount.toFixed(2)}</h2>`);
71 </script>
72 </body>
73 </html>
```



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3. Output:

← → ↺ ⓘ File /Users/shreyamaske/jsexp2/case2.html

☐☐

Shreya's Online Shopping Website

Shopping Invoice

Customer Name: Shreya Maske

Mobile No: 7798642334

Invoice No: 880024

Product: Basmati Rice

Quantity: 6 Kg

Rate per Kg: ₹120

Total: ₹720.00

Discount: ₹14.40

GST (18%): ₹127.01

Packing Charges: ₹35.00

Payment Mode: Card

Membership: No

Final Bill: ₹867.61

4. Result/Conclusion: The online shopping billing webpage was successfully developed using **HTML, CSS, and JavaScript**. The program accepted customer and product details, calculated the total bill including **discount, GST, and packing charges**, and displayed the final invoice correctly. It demonstrated the use of user input, arithmetic calculations, objects, template literals, and object destructuring in JavaScript.



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● Experiment No.: 3

1. **Title:** Implement control structures and form validation; create a grading system based on user-entered marks.

2. **Software/Tools Required:** VS Code, web browser, HTML, JavaScript.

3. **Theory: Control structures** (`if`, `else if`, and `else`) are used to make decisions based on the marks entered by the user.

Form validation checks whether the input is empty or outside the valid range (0–100) before processing.

The `getElementById()` method is used to access the value entered in the input box.

The `Number()` function converts the entered value from text to a numeric value. A **function** (`calculateGrade()`) is called when the **Submit** button is clicked to validate the input and calculate the grade.

The calculated grade is displayed on the webpage using the `innerHTML` property.

4. Output :

The screenshot shows a web browser window with the address bar displaying the file path: `/Users/shreyamaske/jsexp3/Shreya1196exp3.html`. The page has a purple background and features the title "Shreya's 1196 Grading System" in a large, bold, black font. Below the title, the text "Enter Marks:" is displayed. Underneath, there is a text input field with the placeholder text "Enter Marks here". Below the input field is a button labeled "Submit".



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The screenshot shows a web browser window with the title 'Grading System'. The address bar shows the file path: /Users/shreyamaske/jsexp3/Shreya1196exp3.html. The page content includes the heading 'Shreya's 1196 Grading System', a label 'Enter Marks:', a text input field containing '106', and a 'Submit' button. A modal dialog box is displayed over the page, stating 'This page says' and 'Enter Valid Marks between 0 and 100', with an 'OK' button.

The screenshot shows the same web browser window. The modal dialog box now displays 'This page says' and 'Enter Marks', with an 'OK' button. The text input field on the page now contains the placeholder text 'Enter Marks here'.

The screenshot shows the web browser window with the page background changed to light purple. The heading 'Shreya's 1196 Grading System' is at the top. Below it is the label 'Enter Marks:', a text input field containing '74', and a 'Submit' button. The results are displayed below the input field: 'Marks: 74' and 'Grade: B+'.

5. Result/Conclusion: The grading system was successfully implemented using JavaScript. The program validated the user input, assigned grades using if-else control structures, and displayed the correct grade based on the entered marks. It effectively demonstrated the use of control structures and form validation in web development.



॥ वसुधैव कुटुम्बकम् ॥

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Case Study No.: 3

1. **Case Study Title:** Scholarship Eligibility, A college awards scholarships based on marks. Below 50-Not Eligible, 50-69-Bronze, 70-84-Silver, 85-94-Gold, 95-100-Platinum.

Write a JavaScript program that accepts student marks and display the Scholarship category.

Case Study Program Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Scholarship Eligibility</title>
</head>
<body style="background-color:rgba(176, 230, 187, 0.867);">
  <h1>SIT Scholarship Eligibility</h1> <hr><hr>
  <label>Enter Marks:</label><br><br>
  <input type="text" id="marks" placeholder="Enter Marks
here"><br><br>
  <button onclick="calculateEligibility()">Submit</button>
  <h2 id="result"></h2>
  <script>
    function calculateEligibility() {
      let marks=document.getElementById("marks").value;
      if(marks == "")
      {
        alert("Enter Marks");
        return;
      }
    }
  </script>

```



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```
marks = Number(marks);  
if(marks < 0 || marks > 100)  
{  
    alert("Enter Valid Marks for Eligibility");  
    return;  
}  
let grade;  
if(marks >= 95)  
    grade = "Platinum Scholarship";  
else if(marks >= 85)  
    grade = "Gold Scholarship";  
else if(marks >= 70)  
    grade = "Silver Scholarship";  
else if(marks >= 50)  
    grade = "Bronze Scholarship";  
else if(marks <= 50)  
    grade = "Not Eligible for Scholarship";  
document.getElementById("result").innerHTML =  
    "Marks: " + marks + "<br>Grade: " + grade;  
}  
</script>  
</body>  
</html>
```



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3. Output:

Screenshot of the SIT Scholarship Eligibility form in a web browser. The browser title is "Scholarship Eligibility". The address bar shows the file path: /Users/shreyamaske/jsexp3/shreya1196case3.html. The form has a green header with the title "SIT Scholarship Eligibility". Below the header, there is a label "Enter Marks:" followed by a text input field containing "Enter Marks here" and a "Submit" button.

Screenshot of the SIT Scholarship Eligibility form in a web browser with a tooltip. The browser title is "Scholarship Eligibility". The address bar shows the file path: /Users/shreyamaske/jsexp3/shreya1196case3.html. The form has a dark green header with the title "SIT Scholarship Eligibility". Below the header, there is a label "Enter Marks:" followed by a text input field containing "Enter Marks here" and a "Submit" button. A tooltip is visible over the form, displaying the text "This page says Enter Marks" with an "OK" button.

Screenshot of the SIT Scholarship Eligibility form in a web browser showing the result. The browser title is "Scholarship Eligibility". The address bar shows the file path: /Users/shreyamaske/jsexp3/shreya1196case3.html. The form has a green header with the title "SIT Scholarship Eligibility". Below the header, there is a label "Enter Marks:" followed by a text input field containing "32" and a "Submit" button. Below the form, the text "Marks: 32" and "Grade: Not Eligible for Scholarship" is displayed.



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Scholarship Eligibility

File /Users/shreyamaske/jsexp3/shreya1196case3.html



SIT Scholarship Eligibility

Enter Marks:

86

Submit

Marks: 86
Grade: Gold Scholarship

4. Result/Conclusion: The Scholarship Eligibility was successfully implemented using JavaScript. The program validated the user input, assigned eligibility marks using if-else control structures, and displayed the correct Eligibility awards based on the entered marks. It effectively demonstrated the use of control structures and form validation in web development.



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● Experiment No.: 4

1. **Title:** Use function types, scope, and closures; apply try-catch; build a palindrome checker.

2. **Software/Tools Required:** VS Code, JavaScript.

3. Theory:

Anonymous Function: A function without a name, assigned to a variable and used to display the welcome message.

Global Scope: Variables declared outside functions (e.g., language) are accessible throughout the program.

Local Scope: Variables declared inside a function (e.g., topic) are accessible only within that function.

Closure: A function that retains access to its outer function's variables even after the outer function has finished execution. The counter() function demonstrates this by preserving the count variable.

Arrow Function: A concise syntax for writing functions, used here to display output messages.

Try-Catch: Used for exception handling to catch and handle invalid input without stopping program execution.

Palindrome Checker: Compares the original string with its reversed version. If both are equal, the string is a palindrome; otherwise, it is not.

4. Output :

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
pratham@Prathams-MacBook-Air-2 jsexp4 % node "/Users/shreyamaske/jsexp4/shreya1196prac4.js"
=====
Shreya's 1196 Palindrome Checker Program
=====
Language: JavaScript
Topic: Functions
Count: 1
1221 is a palindrome.
```



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```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● pratham@Prathams-MacBook-Air-2 jsexp4 % node "/Users/shreyamaske/jsexp4/shreya1196prac4.js"
=====
Shreya's 1196 Palindrome Checker Program
=====
Language: JavaScript
Topic: Functions
Count: 1
Count: 2
abc1221cba is a palindrome.
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● pratham@Prathams-MacBook-Air-2 jsexp4 % node "/Users/shreyamaske/jsexp4/shreya1196prac4.js"
=====
Shreya's 1196 Palindrome Checker Program
=====
Language: JavaScript
Topic: Functions
Count: 1
daman is not a palindrome.
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● pratham@Prathams-MacBook-Air-2 jsexp4 % node "/Users/shreyamaske/jsexp4/shreya1196prac4.js"
=====
Shreya's 1196 Palindrome Checker Program
=====
Language: JavaScript
Topic: Functions
Count: 1
Count: 2
Error: Input must be a string.
```

5. Result/Conclusion:The program successfully demonstrates the use of JavaScript function types, variable scope, closures, arrow functions, and exception handling using try-catch. It correctly checks whether a given string is a palindrome while illustrating important JavaScript programming concepts in a simple and effective manner.



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Case Study No.: 4

- Case Study Title:** Real time case study ATM card pin verification using JavaScript functions case study an ATM system verifies whether a customers pin is a palindrome. If the pin reads the same forward and backward the Bank displays a special security message. To implement this, create a function to reverse the entered pin number and another function to check whether it is a palindrome. use function declaration, function expression, Arrow function and demonstrate scope and closure concepts.

Case Study Program Code:

```
<!DOCTYPE html>

<html>

<head>

    <title>ATM PIN Verification</title>

</head>

<body style="background-color: rgb(233, 166, 58);">

<h2 >Shreya's (1196) ATM PIN Verification System</h2>

Enter ATM PIN:

<input type="text" id="pin">

<button onclick="verifyPIN()">Verify PIN</button>

<hr>

<div id="output"></div>

<script src="shreya1196case4.js"></script>

</body>

</html>
```



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```
// Anonymous Function (Function Expression)
let welcome = function () {
    return "<h3>ATM PIN Verification System</h3>";
};

// Global Variable
let bankName = "Shreya's (1196) Bank";

// Function Declaration - Scope
function scopeDemo() {
    let branch = "Nagpur"; // Local Variable
    return "<b>Bank:</b> " + bankName +
        "<br><b>Branch:</b> " + branch + "<br>";
}

// Closure
function counter() {
    let count = 0;
    return function () {
        count++;
        return count;
    };
}

let c = counter();

// Function Declaration - Reverse PIN
function reversePIN(pin) {
    return pin.split("").reverse().join("");
}

// Function Declaration - Check Palindrome
function palindrome(pin) {
    let reverse = reversePIN(pin);
    return pin === reverse;
}
```



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```
// Arrow Function
const display = (message) => {
    document.getElementById("output").innerHTML +=
message + "<br>";
};

// Main Function
function verifyPIN() {
    document.getElementById("output").innerHTML =
"";
    display(welcome());
    display(scopeDemo());
    display("<b>Verification Attempt:</b> " + c());
    try {
        let pin =
document.getElementById("pin").value;
        if (pin === "") {
            throw "Please enter ATM PIN. ";
        }
        if (palindrome(pin)) {
            display("<h3 style='color:red'>Weak
Security!</h3>");
            display("PIN is a palindrome.");
            display("Choose a stronger PIN.");
        }
        else {
            display("<h3 style='color:green'>Strong
Security!</h3>");
            display("PIN is not a palindrome.");
        }
    }
    catch(error) {
        display("<b>Error:</b> " + error);
    }
}
```



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}

3. Output:

24070521196_Shreya Maske_ x

ATM PIN Verification x +

< > ↻

File /Users/shreyamaske/jsexp4/shreya1196case4.html

⌵

Shreya's (1196) ATM PIN Verification System

Enter ATM PIN: Verify PIN

ATM PIN Verification System

Bank: Shreya's (1196) Bank
Branch: Nagpur

Verification Attempt: 1

Strong Security!

PIN is not a palindrome.



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24070521196_Shreya Maske

ATM PIN Verification

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File

/Users/shreyamaske/jsexp4/shreya1196case4.html

⌵

Shreya's (1196) ATM PIN Verification System

Enter ATM PIN:

ATM PIN Verification System

Bank: Shreya's (1196) Bank
Branch: Nagpur

Verification Attempt: 2

Weak Security!

PIN is a palindrome.
Choose a stronger PIN.

4. Result/Conclusion:

The ATM PIN verification system was successfully implemented using JavaScript. The program correctly identified whether the entered PIN was a palindrome, displayed "**Weak Security**" for palindrome PINs and "**Strong Security**" for non-palindrome PINs, while demonstrating function declaration, function expression, arrow function, scope, closures, and exception handling using try-catch.